



Atmospheric Arithmetic: Evaluating Climate Mitigation Strategies

Abstract

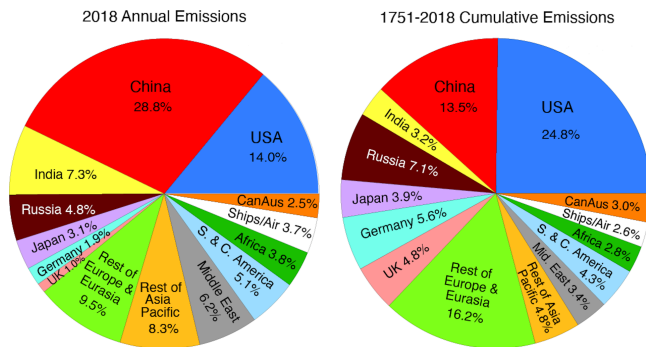
When politicians promise net-zero emissions, they often count on trees or technology to pull carbon dioxide out of the air. In 2022, climate scientist Kevin Trenberth did the math. He showed that removing just 1 part per million of carbon dioxide would cost hundreds of billions of dollars, and concluded that there is no viable workaround to cutting fossil fuel emissions.

Introduction

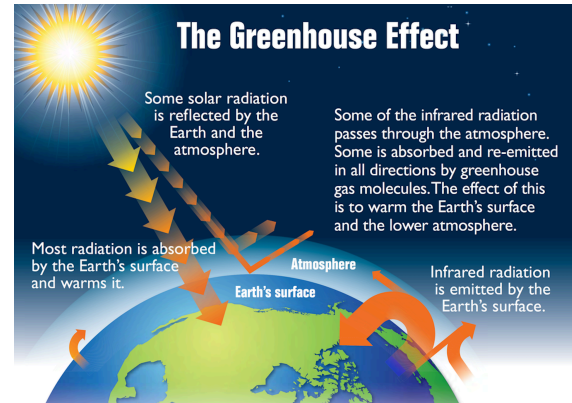
Burning fossil fuels releases carbon dioxide, which stays in the atmosphere for centuries and acts like a blanket that traps energy near Earth's surface. The planet has warmed an estimated 1.1°C (2°F) since about 1880 and is on its way to 1.5°C, the limit the Paris Agreement hoped not to cross. Studies show emissions must reach net zero by midcentury to have a chance of limiting warming even to 2°C. China is currently the largest emitter, but accumulated emissions matter most, and there the United States leads. The most effective solution is to decarbonize economies with renewable energy and electric vehicles. Yet many leaders turn to alternatives instead.

How Do Scientists Study This?

FROM THE ARTICLE

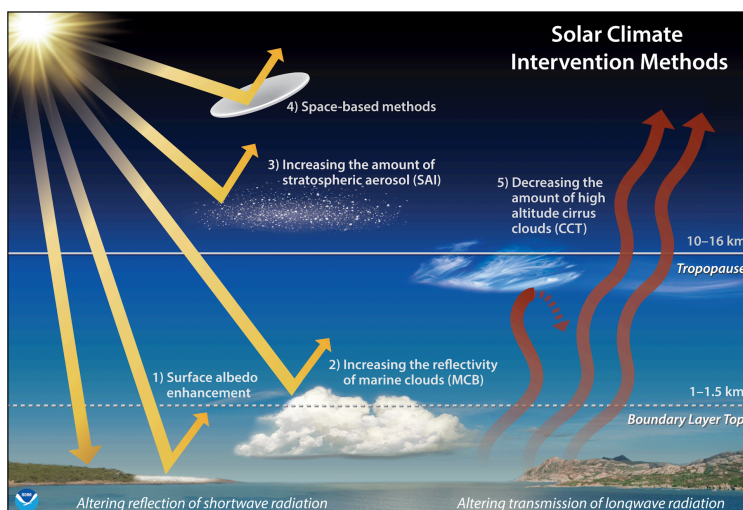


Shares of carbon dioxide emissions from fossil fuels in 2018 compared with cumulative emissions, 1751-2018. Kevin Trenberth, based on BP data.



How the greenhouse effect works. Image: EPA.

Trenberth evaluated four popular alternatives to cutting emissions: solar geoengineering, carbon capture and storage, direct air capture, and tree planting. For each, he considered how it works and its risks. Then he converted everything into the same units. Carbon dioxide at Mauna Loa, Hawaii, is measured in parts per million by volume, but carbon removal is priced in metric tons. Using the mass of the atmosphere, about 5.5 quadrillion metric tons, he calculated that 1 part per million of carbon dioxide equals about 7.8 billion metric tons. He then multiplied by cost estimates from the World Resources Institute to find the price of removing 1 ppm.

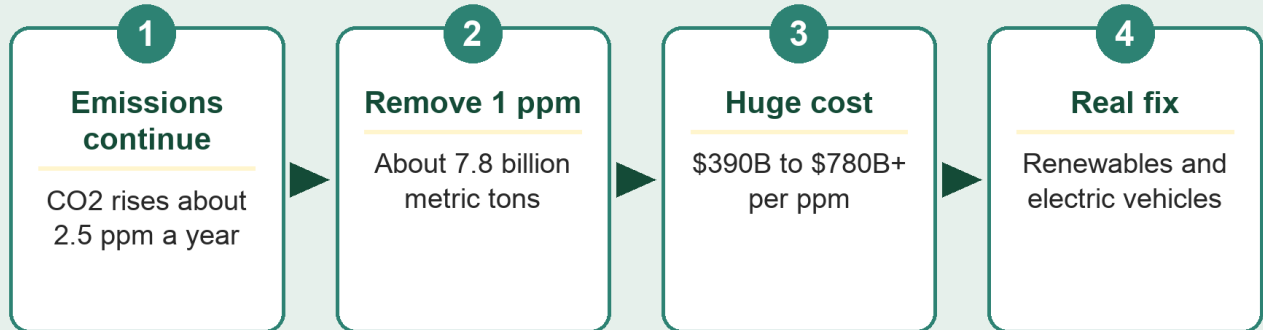


FROM THE ARTICLE

Proposed methods of solar radiation management. Image: Chelsea Thompson, NOAA/CIRES.

Explore the Evidence

Why removal can't replace cutting



List the options

Examine geoengineering, carbon capture, direct air capture, and tree planting.

Convert the units

1 part per million of CO2 in the atmosphere equals about 7.8 billion metric tons.

Calculate the cost

Multiply tons by cost per ton to find the price of removing 1 ppm.

What Did They Discover?

The yearly rise in carbon dioxide has accelerated from about 1 ppm in the 1960s to 1.5 in the 1990s and 2.5 since 2010, even through the pandemic. Direct air capture costs \$250 to \$600 per ton; even at \$100 per ton, removing 1 ppm would cost about \$780 billion. Tree restoration could remove about 7.4 gigatons by 2050, less than 1 ppm, at up to \$50 per ton, or as much as \$390 billion per ppm, and trees can burn, decay, or die. Only about a dozen U.S. industrial plants capture their carbon, mostly to enhance oil drilling. Solar geoengineering, which mimics a volcano's cooling, could disrupt rainfall, as the 1991 Pinatubo eruption did.

Atmospheric arithmetic

7.8 Gt of CO₂ equals 1 part per million in the air

\$780B to remove 1 ppm by direct air capture at \$100/ton

~420 ppm of CO₂ today, up from about 280

Carbon dioxide has risen from about 280 ppm before industrialization to around 420 ppm and is still rising more than 2 ppm per year.

Takeaways

Do the math

Removing 1 ppm of CO₂ costs hundreds of billions of dollars.

Workarounds fall short

Trees, capture, and geoengineering make only small dents or carry risks.

Cut emissions

Switching to renewable energy is the only viable solution.

Key Terms

Net zero

Balancing the greenhouse gases put into the atmosphere with the amount removed, so there is no net increase.

Direct air capture

Technology that pulls carbon dioxide directly from the air, using large amounts of energy.

Carbon capture and storage

Capturing carbon dioxide from industrial sources and storing it underground.

Solar geoengineering

Deliberately reflecting sunlight away from Earth, for example by adding particles to the stratosphere.

Parts per million (ppm)

A unit for small concentrations; 1 ppm of CO₂ is one molecule of CO₂ per million molecules of air.

Words to Know · Atmospheric Arithmetic

Net zero

Direct air capture

Carbon capture and storage

Solar geoengineering

Parts per million (ppm)

Activity

The Price of 1 ppm

Students repeat Trenberth's atmospheric arithmetic to compare the costs of carbon removal strategies.

Materials: Calculator; Cost table from the lesson; Graph paper

1. Start with the conversion: 1 ppm of CO₂ = 7.8 billion metric tons.
2. Calculate the cost of removing 1 ppm with direct air capture at \$100, \$250, and \$600 per ton.
3. Calculate the cost with trees at \$50 per ton.
4. CO₂ is rising about 2.5 ppm per year. How much would it cost just to cancel one year of increase with each method?
5. Make a bar graph of your results and write a conclusion about the best strategy.

References

Trenberth, K. (2022). How not to solve the climate change problem. The Conversation.
<https://doi.org/10.64628/AAI.eyu4r7dch>